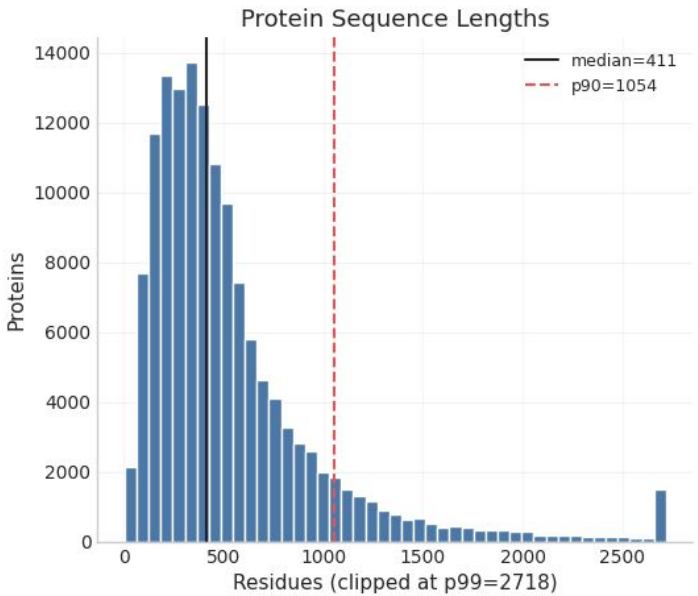
Sequence-based baseline + AlphaFold structural information

“Whether **structural information** can improve function prediction beyond **sequence only features**.?”

Data aqusition

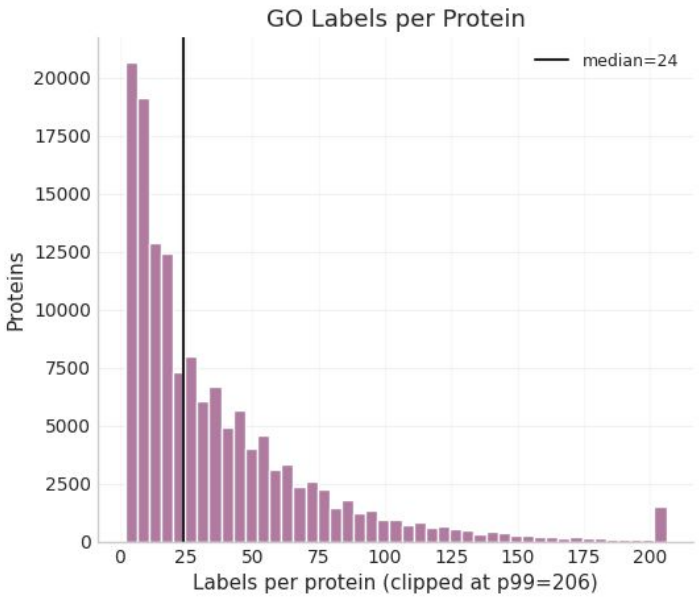
Three Challenges

1. Protein sequences greatly vary in length.



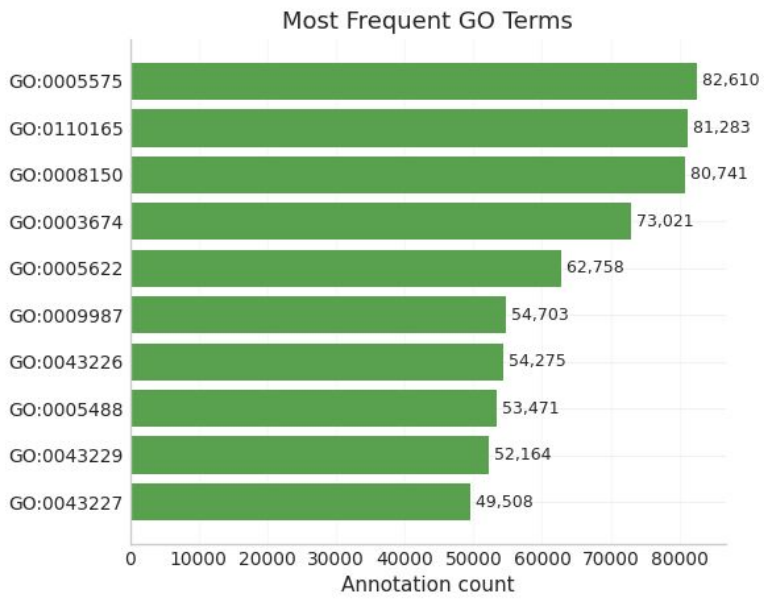
Three Challenges

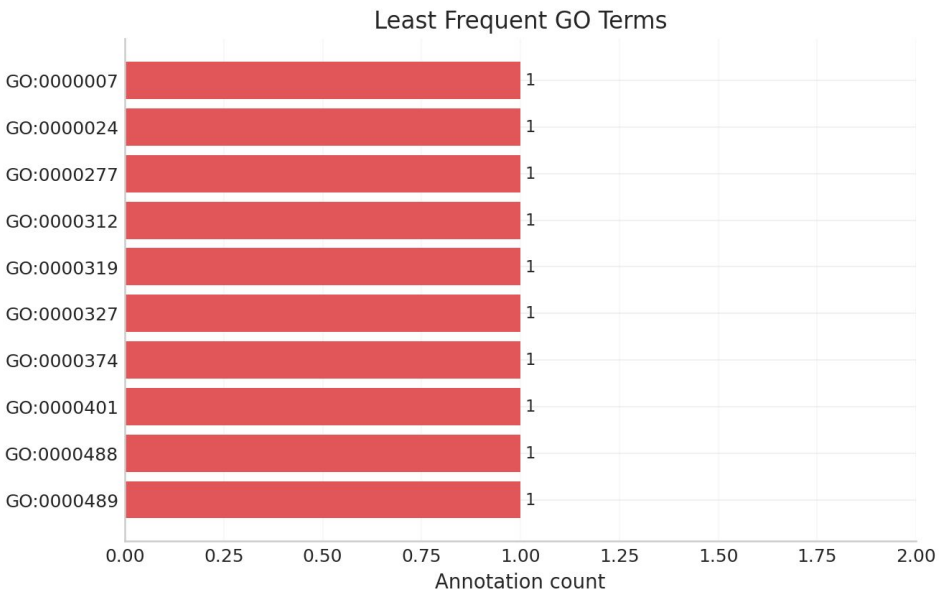
2. GO label space is very large.



Three Challenges

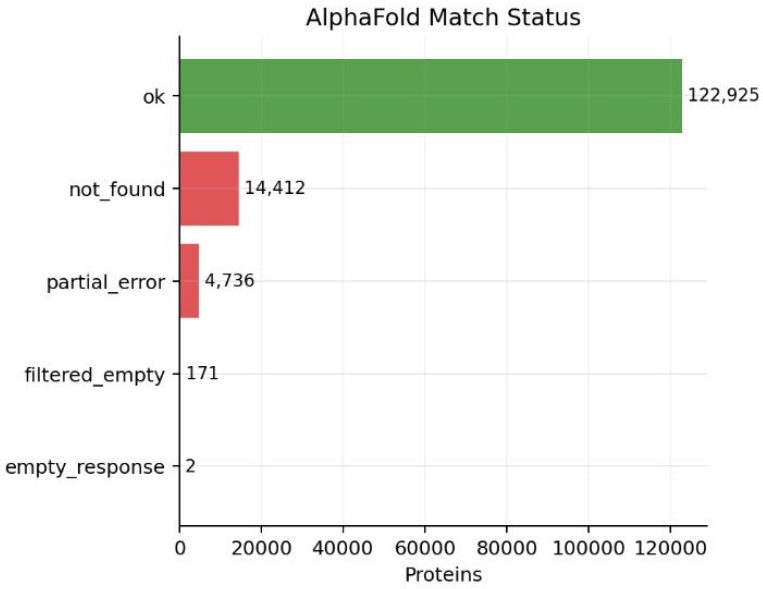
3. GO term frequencies are highly long-tailed.

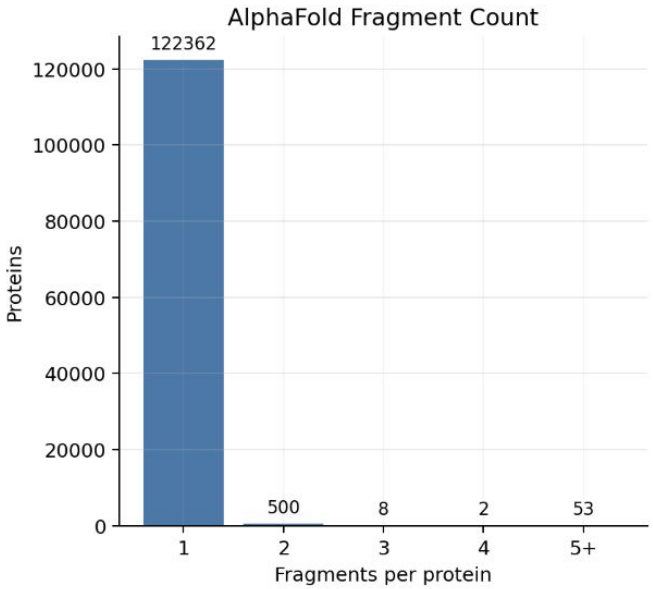




Data prepare - Download Alphfold PDB

AlphaFold Protein Structure Database

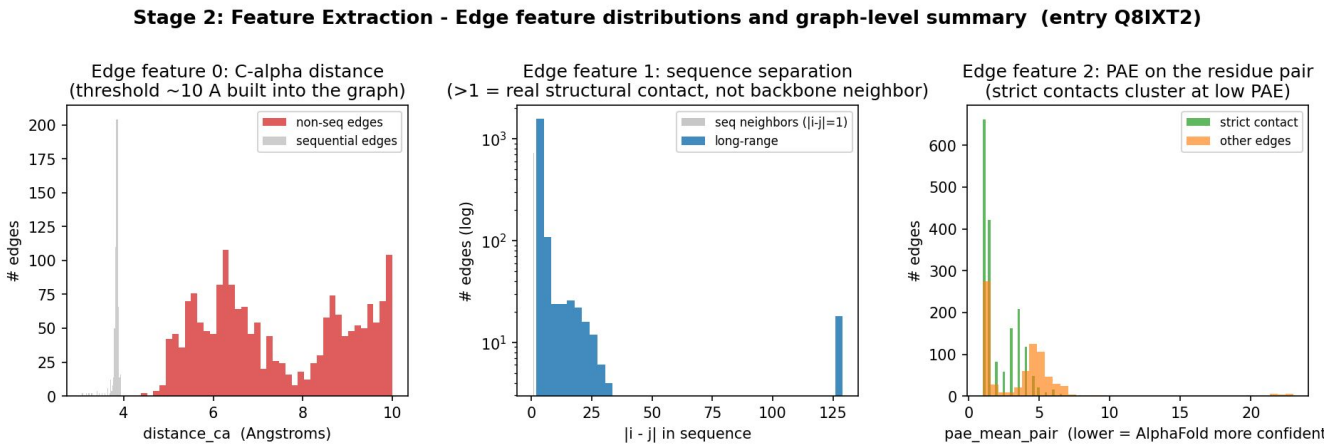




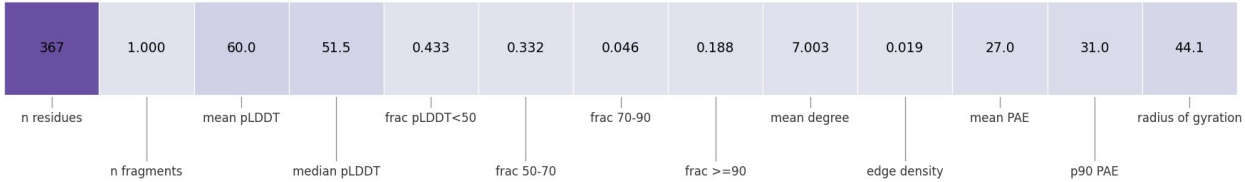
Data prepare -Turn protein structure into graph representation

One residue -> 32-dim base feature vector (example: residue idx 83, AA = 'Q', pLDDT = 98.8, contact_degree = 12)

Data prepare -Turn protein structure into graph representation

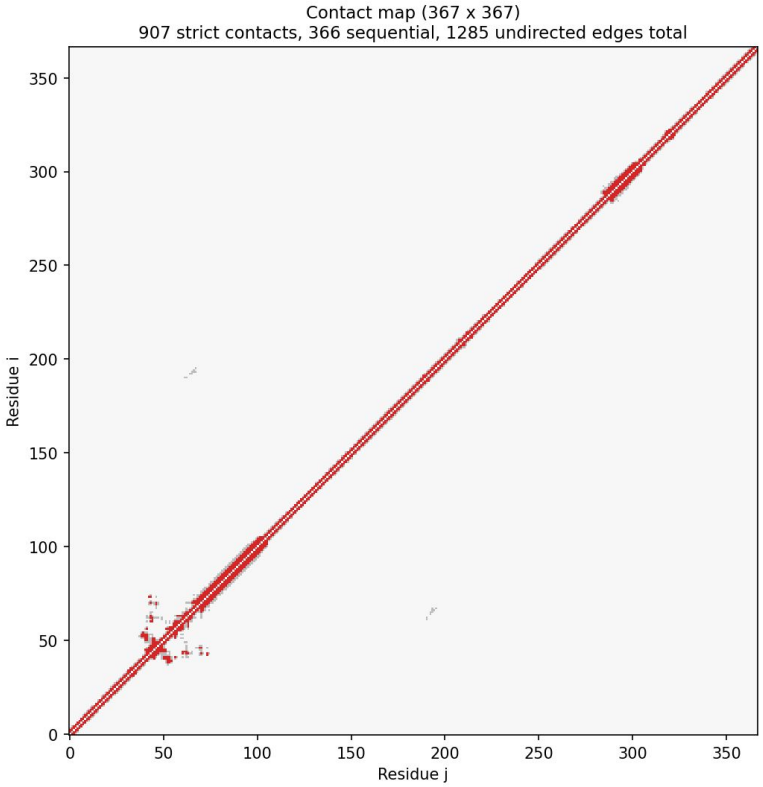


Graph-level feature vector graph_feat: shape (13,) - one summary vector per whole protein



Data prepare -Turn protein structure into graph representation

Stage 2: Feature Extraction - Edge side / contact map (entry Q8IXT2)

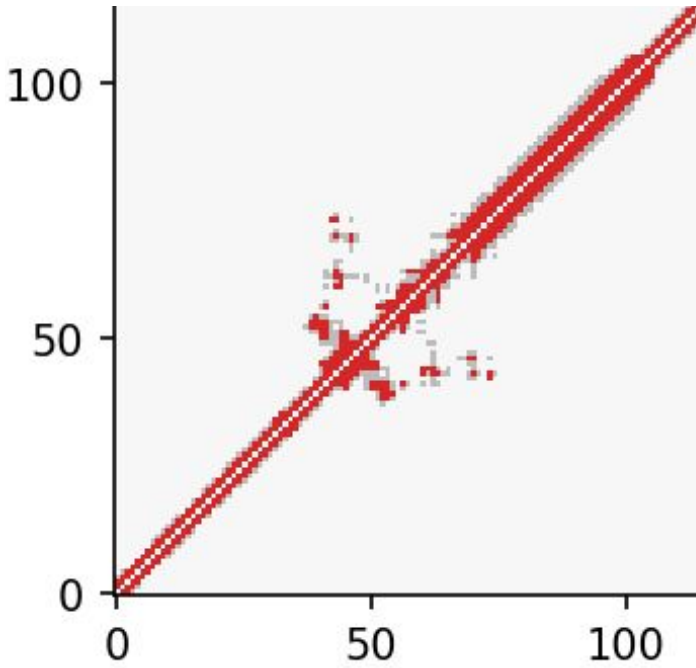


Edge type

sequential / weak contact
(usually backbone neighbors)

strict 3D contact
(< distance threshold)

Reading the map
- diagonal: residue i ~ i+1
  (sequential backbone)
- off-diagonal red dots:
  long-range 3D contacts
  (the structural signal
  that sequence alone
  cannot see)

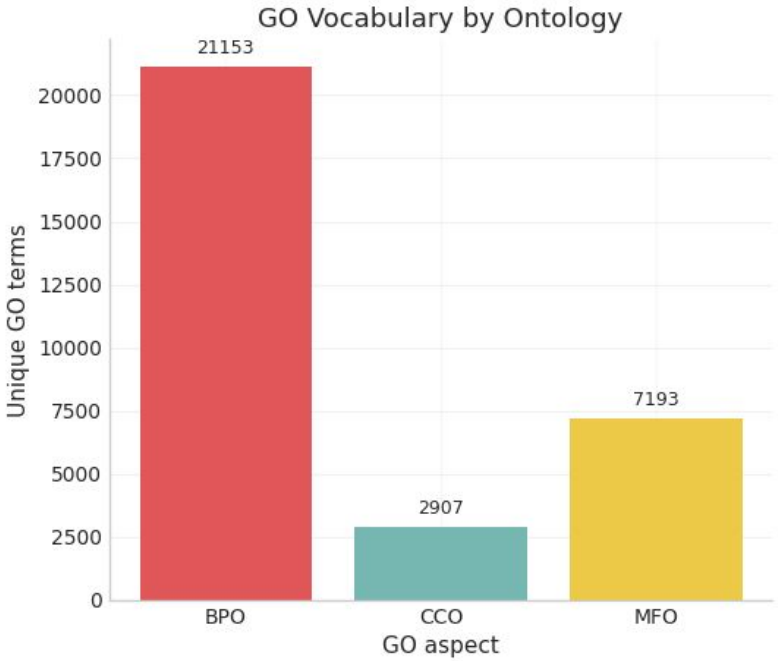


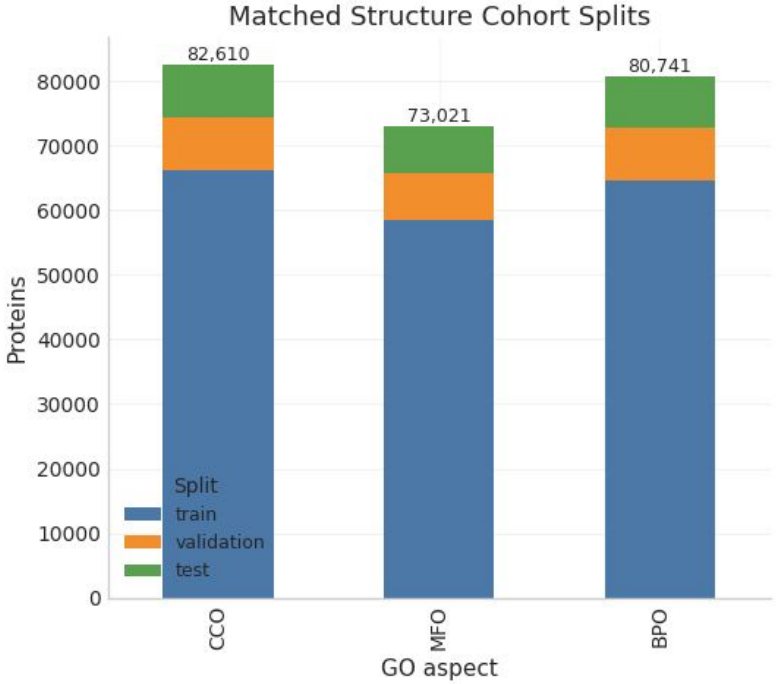
Node, Edge, and Graph data in the protein structure-graph dataset

what each tensor describes, its shape, and the kind of features it stores

feature aspect	**Node data** *per-residue*	**Edge data** *per-contact*	**Graph data** *per-protein*
Attached to	each residue (one amino acid)	each residue-residue contact (one edge)	the whole protein (not residues, not contacts)
Tensor shape (per protein)	x : (N, 682)	edge_index : (2, E) edge_attr : (E, 6)	graph_feat : (13,)
Count per protein	N (= sequence length)	E (roughly ~10 x N for our graphs)	1 (single fixed-size vector)
Example features	amino-acid one-hot, pLDDT + bucket flags, PAE row stats, contact degree, position fraction (i / L)	C-alpha distance, sequence separation |i - j|, PAE on this pair, strict-3D-contact indicator, backbone-vs-non-seq flag	protein length, fragment count, proportion of residues in each pLDDT bucket, mean PAE, edge density, radius of gyration, ...

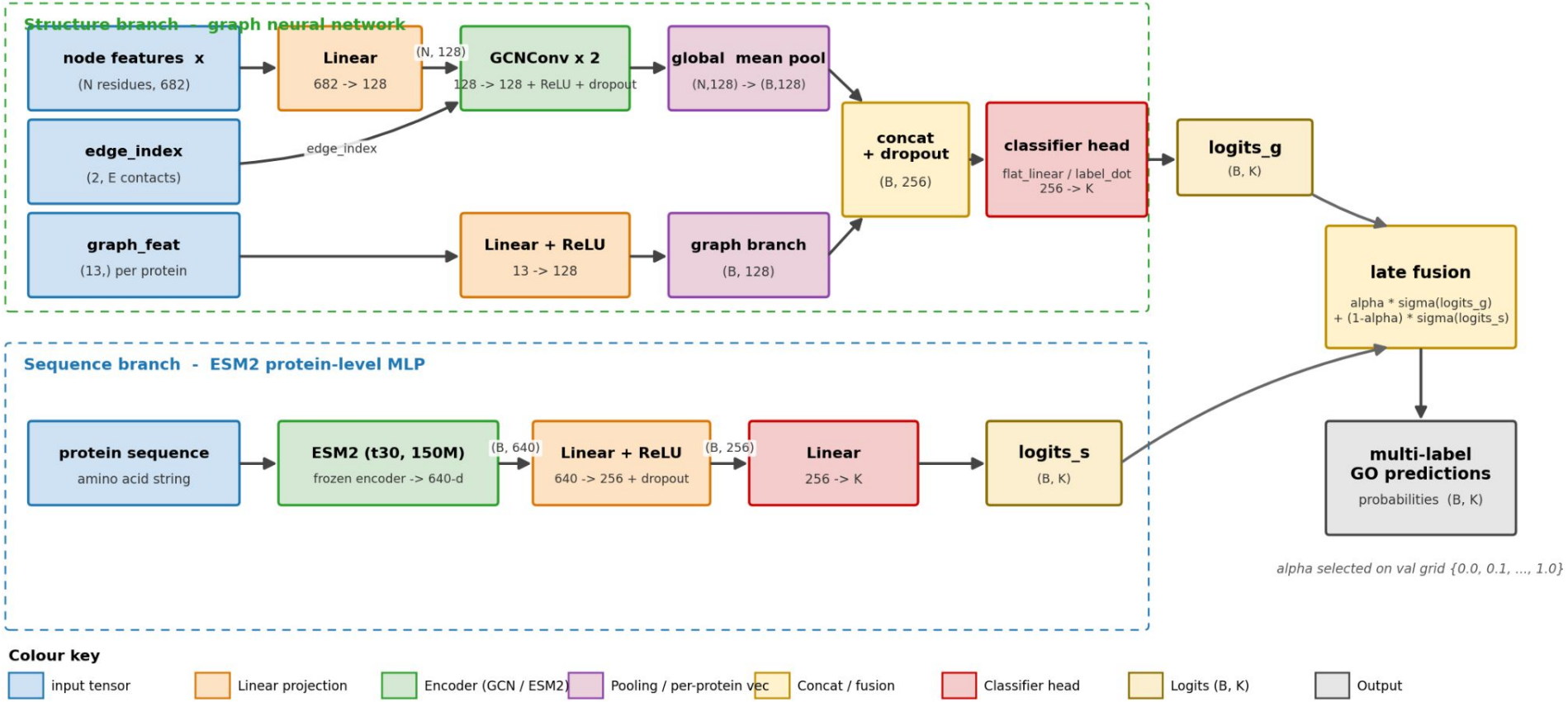
Data set and split



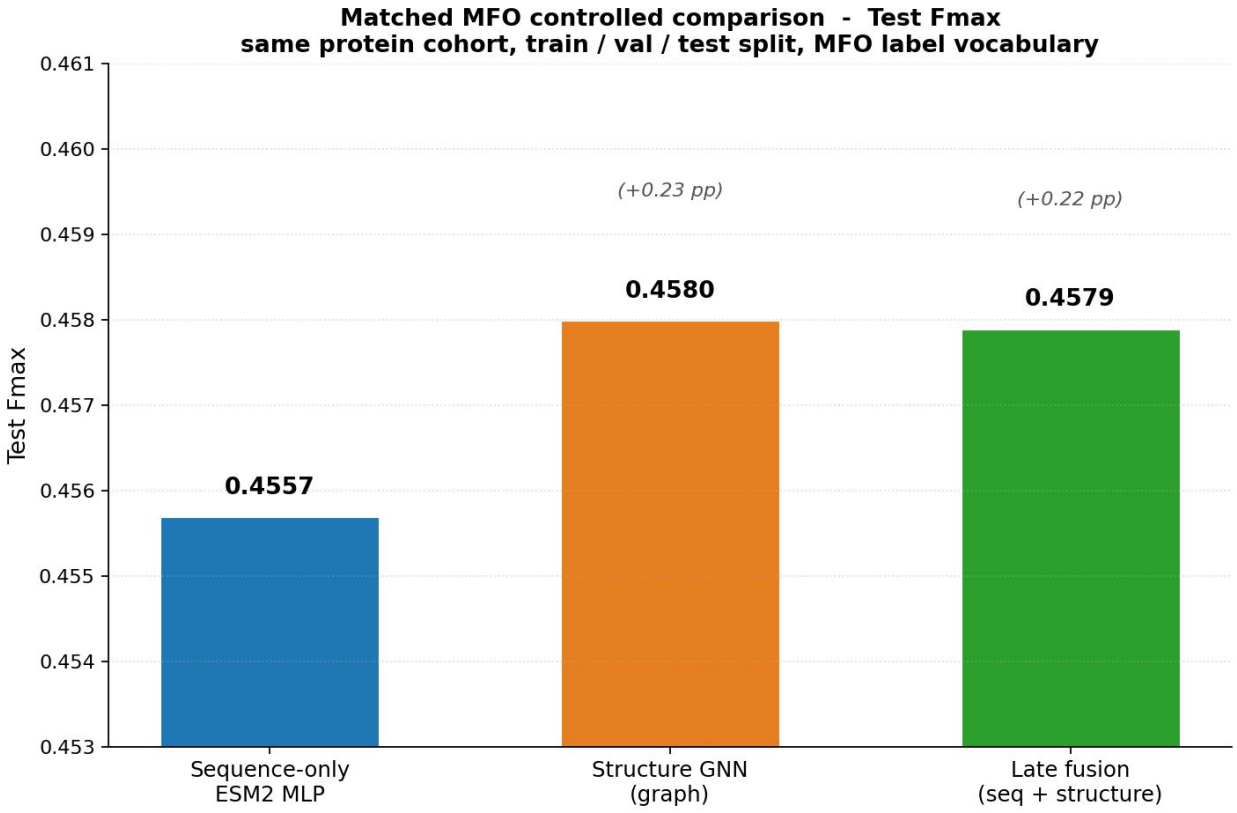


Late-fusion GO classifier (structure GNN + sequence MLP)

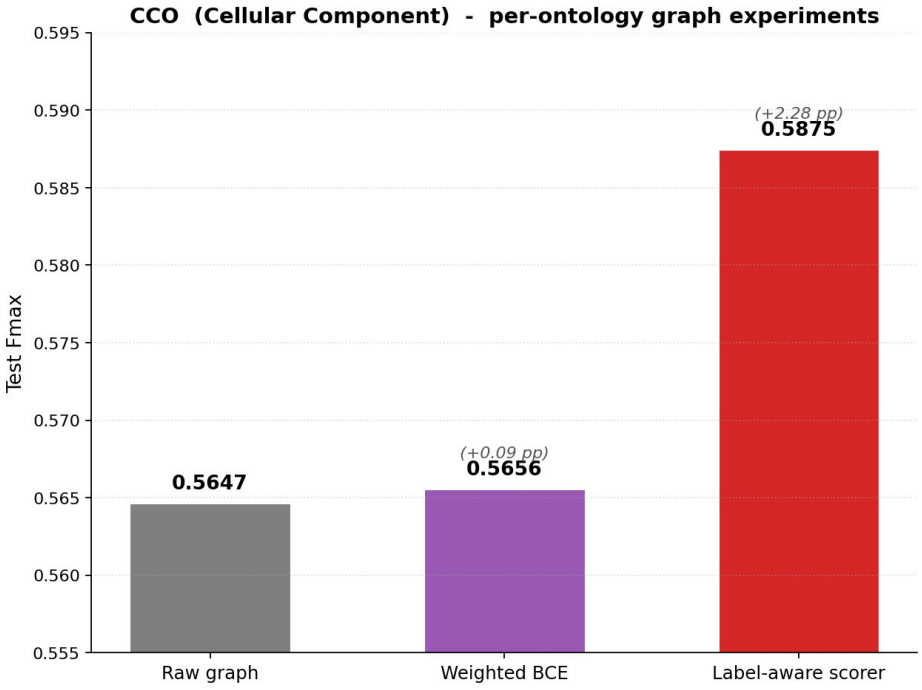
*two parallel branches produce per-label scores; fusion = alpha * sigma(logits_g) + (1 - alpha) * sigma(logits_s)*

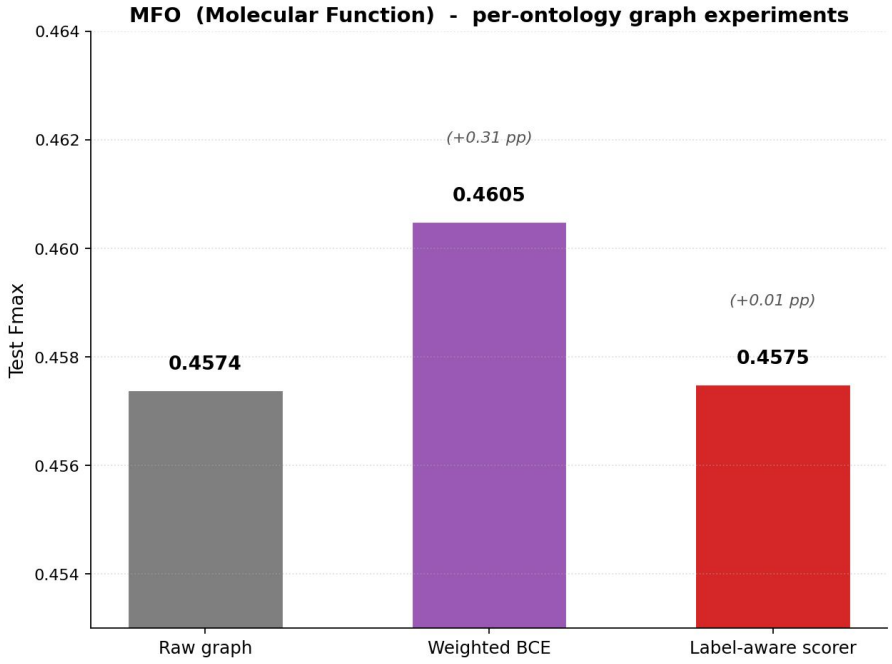


Training result



Training result





Conclusion